b.grad = ? $\quad \frac{\partial L}{\partial b} = ?$, $n = (x_1 w_1 + x_2 w_2) + b$

chain rule

$$\frac{dz}{dx} = \left(\frac{dy}{dx}\right) * \left(\frac{dz}{dy}\right)$$

$$\frac{\partial L}{\partial b} = \left(\frac{\partial n}{\partial b}\right) * \left(\frac{\partial L}{\partial n}\right)$$

$$b.grad = \left(\frac{\partial n}{\partial b}\right) * n.grad$$

$\frac{\partial n}{\partial b} = ?$. A) slope = ?

$$\frac{f(b+h) - f(b)}{h} = \frac{\left(\cancel{x_1 w_1 + x_2 w_2}\right) + (\cancel{b}+h) - \left(\cancel{(x_1 w_1 + x_2 w_2)} + \cancel{b}\right)}{h} = 1$$

$$\boxed{\frac{\partial n}{\partial b} = \text{local der.} = 1}$$

b.grad = n.grad ☆☆

on, observation, when a add^n happens the local derivative turns out to be equal to 1.

(+) → gradient remain Same

b.grad = n.grad.

★★★

if b.grad = n.grad,

so, from above obs, we can say

$$\boxed{(x_1w_1 + x_2w_2).grad = b.grad = \underline{n}.grad = 0.5}$$

$x_1w_1 + x_2w_2 = x_1w_1 + x_2w_2$

$x_1w_1.grad = ?$, $x_2w_2.grad = ?$

From above property, as add^n happens,

so, $\boxed{(x_1w_1 + x_2w_2).grad = x_1w_1.grad = x_2w_2.grad}$ $= 0.5$

$x2w2 = x2 * w2$

$$\frac{\partial L}{\partial x2} = ?$$

$$\frac{\partial L}{\partial x2} = \frac{\partial (x2w2)}{\partial x2} * \frac{\partial L}{\partial (x2w2)}$$

$$x2.grad = \left(\frac{\partial (x_2 w_2)}{\partial x_2}\right) * x2w2.grad$$

$$\frac{\partial (x_2 w_2)}{\partial x2} = ?$$

$$slope = \frac{f(x+h) - f(x)}{h}$$

$$= \frac{(x2+h)w2 - (x2w2)}{h}$$

$$= \left(\cancel{x2w2} + \cancel{h}w2 - \cancel{x2w2}\right) / \cancel{h}$$

$$\boxed{= w2} \rightarrow \text{local der.}$$

Analytic

$$\boxed{x2.grad = w2 * (x2w2).grad}$$

As per obs. if '*' comes, then multiply with the opp. term to which grad. has to be calc. with the global derivative, as opp. term/data is the local der.

$$c = a * b$$

$$a.grad = b * c.grad$$

so,

$$x2.grad = (w2) * (x2w2).grad$$

$$w2.grad = (x2) * (x2w2).grad$$

$$x2.grad = 1 * 0.5 = 0.5$$

$$w2.grad = 0 * 0.5 = \cancel{0.5}\ 0$$

$$x1.grad = (w1) * (x1w1).grad$$

$$w1.grad = (x1) * (x1w1).grad$$

$$x1.grad = -3 * 0.5 = -1.5$$

$$w1.grad = 2 * 0.5 = 1$$